Transfer: energy and work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose and justify energy methods in new physical situations.

SUCCESS CRITERIA - I can...

1. Choose the correct energy terms and units.
2. Show a consistent numerical balance.
3. Explain what your calculation treats as true and when it stops working.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “work” and “power” to explain your answer. Sentence starters are optional.

work: energy transferred by a force along a displacement

power: rate of energy transfer

energy budget: an account of all energy stored and transferred

01 Warm-up: bring back an earlier idea

At fixed speed, mass becomes **3** times as large while energy-coupling fraction becomes **1/5** as large. **Compare coupled energy**.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $W_{net} = \Delta K$; $K = \frac{1}{2}mv^2$; $U_g = mgh$; $U_s = \frac{1}{2}kx^2$
$P = W/\Delta t$.
g = **10 N/kg** where needed. Select relationships for the system and facts stated here.

Worked reasoning: A **2 kg** cart drops **2 m** with g =**10**, then loses **10 J** to friction. From rest, its final kinetic energy is **40**-**10**=**30 J**, so speed √**30**≈**5.48 m/s**.

Your turn: Draw a complete energy account and a sketch identifying where friction acts.

Sentence starter: In my work, I used … to show … . I labeled it … .

Initial | **Transfers** | **Final**

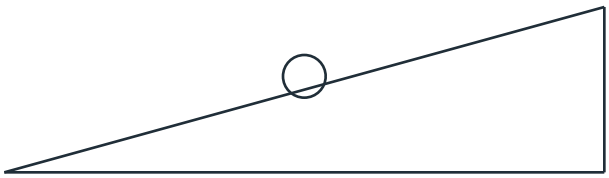
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A cart coasts down a frictionless ramp from rest. **Explain which energy decreases** and which increases. **Mark the vertical height** in the sketch.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



The marker represents the object on the sloping surface.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A **2 kg** cart starts from rest **3 m** above the bottom of a ramp. It reaches the bottom at **4 m/s**, then compresses a **200 N/m** spring on a frictionless level section. Use g = **10 N/kg**. **Complete the energy account** to **find the ramp's energy loss** and maximum spring compression.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A motor supplies **100 J** while the load gains **70 J** mechanical energy. **Explain where the other 30 J** might go and whether energy was destroyed.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A ball is thrown vertically upward at **20 m/s** from h=**0**. Use g=**10** and neglect drag. **Find maximum height** by energy.
Show the energy changes and transfers; **label units**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**

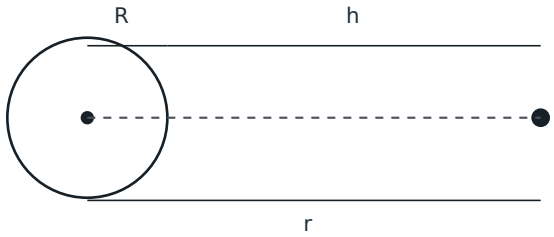
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

A simulated source has GM=**200 m³/s²**. At r=**4 m**, find escape speed and **name two things your calculation** treats as true. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **3 kg** cart starts at **2 m/s**, drops **1 m**, then compresses a spring k=**200 N/m**. Friction removes **11 J** before maximum compression; g=**10**.
1. **Find compression** and **check the final energy account**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3